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Project Planning Institute

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Month 1

Project Controls Foundations & Scope Planning

Duration	4 weeks

Month Journey

- **Week 1:** Project lifecycles, stakeholders, project-controls roles, governance and terminology
- **Week 2:** Scope definition, deliverables, work breakdown structure (WBS), coding and assumptions
- **Week 3:** Activities, dependencies, durations, milestones and logic relationships
- **Week 4:** Hand-calculated CPM, float, schedule narrative and Excel Gantt/WBS lab

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Month 2

Primavera P6, Microsoft Project & Scheduling

Duration	4 weeks

Month Journey

- **Week 1:** Set up P6 enterprise/project structures, calendars, activity codes and WBS
- **Week 2:** Build logic-driven schedules and identify the critical path in P6
- **Week 3:** Recreate, baseline and update a schedule in Microsoft Project; compare platforms
- **Week 4:** Advanced float, constraints, PERT, what-if analysis and schedule optimisation

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Month 3

Resource Planning, Budgeting & Cost Control

Duration 4 weeks

Month Journey

- **Week 1:** Resource breakdown structures, labour/plant/material assignments and productivity
- **Week 2:** Resource histograms, overload detection, levelling and capacity scenarios
- **Week 3:** Estimates, cost codes, budgets, cost loading, cash flow and S-curves
- **Week 4:** Integrated schedule–cost reconciliation using Excel formulas, pivot tables and charts

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Month 4

Earned Value & Progress Measurement

Duration 4 weeks

Month Journey

- **Week 1:** Earned value: PV, EV, AC, CPI, SPI, BAC, EAC and ETC

Abbreviation	Theme	Primary Capability
PV	Planned Value	The budgeted cost of work scheduled to be completed by a particular date.
EV	Earned Value	The budgeted value of work actually completed.
AC	Actual Cost	The actual amount spent on work completed.
CPI	Cost Performance Index	Measures how efficiently the project is using its budget.
SPI	Schedule Performance Index	Measures project progress against the planned schedule.
BAC	Budget at Completion	The total approved budget for the entire project.
EAC	Estimate at Completion	The forecasted total cost of the project when completed.
ETC	Estimate to Complete	The estimated additional cost required to finish the remaining work.

- **Week 2:** Measurement rules, progress quantities, physical percent complete and cut-off dates
- **Week 3:** Collect contractor updates using controlled Excel templates and validation logs
- **Week 4:** Rebaseline rules, schedule updates, variance analysis and executive exception reports

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Month 5

Schedule Quality, Risk & Forecasting

Duration	4 weeks

Month Journey

- **Week 1:** Schedule integrity checks, logic quality, constraints, float and health metrics
- **Week 2:** Risk identification, risk registers, probability/impact and mitigation owners
- **Week 3:** Weather allowances, calendars, uncertainty, PERT and quantitative schedule risk
- **Week 4:** Build a risk-adjusted forecast and present confidence ranges and mitigations

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Month 6

Delay Analysis, Change & Recovery

Duration 4 weeks

Month Journey

- **Week 1:** Delay event records, contemporaneous evidence and critical-path impacts
- **Week 2:** Fragnets, time-impact analysis, windows analysis and forensic principles
- **Week 3:** Change control, notices, contractual interfaces and claims-support narratives
- **Week 4:** Recovery options: resequencing, crashing, fast-tracking and scenario comparison

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Month 7

Contracts, Procurement & Commissioning

Duration 4 weeks

Month Journey

- **Week 1:** Contract milestones, long-lead items and vendor data requirements
- **Week 2:** Procurement logic: requisition, approval, purchase order, manufacture and delivery
- **Week 3:** Create and operate Excel procurement tracker, expediting and interface register
- **Week 4:** Testing, handover, commissioning, punch lists and integrated completion schedule

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Month 8

Lean Planning, Collaboration & Reporting

Duration 4 weeks

Month Journey

- **Week 1:** Lean principles, Last Planner, lookahead planning and constraint removal
- **Week 2:** Weekly work plans, percent plan complete and coordination meetings
- **Week 3:** Project-controls report architecture: data model, KPIs and schedule narratives
- **Week 4:** Import Excel and schedule exports into Power BI; create measures and visual dashboards

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Month 9

Power BI, Tableau & Digital Project Controls

Duration 4 weeks

Month Journey

- **Week 1:** Power Query cleaning, relationships, star schemas and refresh workflow
- **Week 2:** DAX measures, earned-value KPIs, variance indicators and drill-through
- **Week 3:** Tableau data connections, calculations, dashboard design and storytelling
- **Week 4:** Data governance, dashboard QA, responsible AI support and executive presentation

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Month 10

Program & Portfolio Controls

Duration 4 weeks

Month Journey

- **Week 1:** Multi-project resource conflicts, shared calendars and programme milestones
- **Week 2:** Portfolio prioritisation, aggregated cost/schedule risk and reporting cadence
- **Week 3:** Apply controls to commercial construction, infrastructure and oil & gas scenarios
- **Week 4:** Apply controls to energy, healthcare, transport and industrial scenarios

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Month 11

Industry Applications & Data Center Deep Track

Duration	4 weeks

Month Journey

- **Week 1:** Sector studio: select contrasting projects and document planning assumptions
- **Week 2:** Data center design, procurement, construction, MEP, testing and commissioning logic
- **Week 3:** Integrate schedules, procurement tracker, EVM, risk register and BI reporting
- **Week 4:** Peer review and client-style defence of sector-specific schedule scenarios

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Month 12

Capstone, Professional Defence & Career Readiness

Duration 4 weeks

Month Journey

- **Week 1:** Capstone brief, project controls plan, WBS, integrated baseline and controls setup
- **Week 2:** Run a simulated reporting period with changes, risks, delay and recovery
- **Week 3:** Publish final Excel, Power BI and Tableau outputs; draft executive narratives
- **Week 4:** Final professional defence, portfolio review, CV/interview coaching and certification action plan

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